A second look, measured

Day 09: a cross-encoder reranks the hybrid's candidates.

TWO WAYS TO SCORE A CHUNK

bi-encoder (dense, Day 04)

question → vector

chunk → vector

made apart, then compared
fast: chunk vectors made in advance

cross-encoder (today)

question + chunk

↓

one score

read together, word by word
slow: one model run per pair

SAME THREE CHUNKS, ANOTHER ORDER

"How much time off do new dads get?"

dense	cross-encoder
#1 Returning to work	#1 Leave entitlement ✓
#2 Who is eligible	#2 Who is eligible
#3 Leave entitlement ✓	#3 Returning to work

RULES COMMITTED BEFORE THE TEST SPLIT

	reranked vs hybrid alone
dev questions ranked first	15 vs 13
test questions ranked first	13 vs 14
reranking for later days	not adopted

one test question decided it; the criterion was written down first

What moved, and why

Every rank the reranker changed, in three groups.

IT READS QUESTION AND CHUNK TOGETHER

"How long is parental leave?"
it meets "16 weeks"
#2 → #1 ✓

RRF's 10 demoted lookups ("USD 220")
the chunk holding the value goes first
#2-3 → #1 ✓

"How much time off do new dads get?"
reranking RRF's top 20
#15 → #1 ✓

IT SCORES EACH CHUNK ALONE

"How soon must I report a lost laptop?"
"How to report" never says laptop · RRF, top 20
#2 → #5 ×

IT DOESN'T REASON

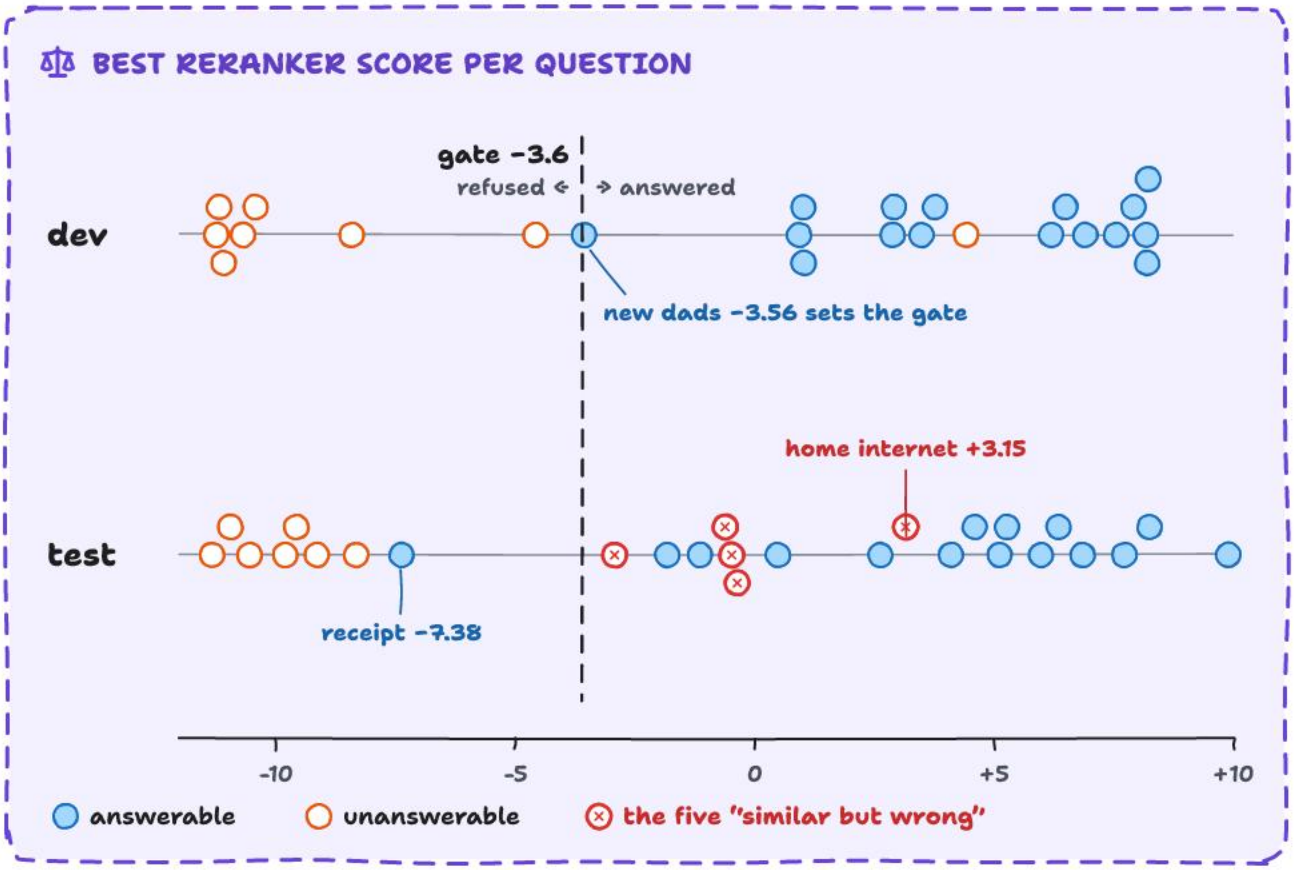
"Receipt for a USD 40 dinner?"
the rule: "over USD 25"
#1 → #2 ×

"Wine with a business dinner?"
the chunk says "Alcohol" · RRF, top 20
#3 → #4 ×

the last two need more than a reranker: context from another section, and a reader that knows 40 > 25 and wine is alcohol

Close isn't answers

Each question's best reranker score, and a gate set on dev.



THE PRE-REGISTERED TEST, ON THE TEST SPLIT

the five: 0 refused

answerable: 1 refused (the receipt question)

a web-search reranker judges "about", not "answers"

not adopted as a gate; answers are checked on Day 12

Before you add a reranker

Three checks and the decision, from today's measurements.

EVERY CANDIDATE IS A MODEL RUN

embed the question	~10 ms
rerank 3	~30 ms
rerank 20	~240 ms

SELECTING THE CONTEXT

- ✓ a copy under another title: dropped
- × an old version, USD 400 not 500: kept (dedupe isn't freshness)
- × one chunk per document: 3 answers lost

IT PROMOTES WHAT IT'S GIVEN

one unfiltered retriever: a managers-only chunk in an employee's pay question

filter every retriever, then rerank



DECISIONS, CRITERIA SET IN ADVANCE

rerank the hybrid's top 3: 13 vs 14 first on test

a gate on its score: 1 answerable refused, 0 of the 5

both: not adopted; the weighted fusion stays

next: Day 10, query transformations